Name: ____________________________ Date: ________________ Period: _________

Spiral Review — Key

1. Recall the parametrization of the unit circle $x^2 + y^2 = 1$. Fill in the component functions and domain.

$x(t) = \cos t$ $y(t) = \sin t$ domain: $0 \le t \le 2\pi$

2. For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find the x-intercepts and y-intercepts.

x-intercepts: $(\pm 5, 0)$ y-intercepts: $(0, \pm 3)$

3. Let $x(t) = 5\cos t$ and $y(t) = 3\sin t$. Complete the table, then verify that one of the points satisfies $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

t	$x(t)$	$y(t)$	Point (x, y)
0	5	0	(5, 0)
$\pi/2$	0	3	(0, 3)
π	-5	0	$(-5, 0)$
$3\pi/2$	0	-3	$(0, -3)$

Verify (using $t = \pi/2$, point $(0, 3)$): $\frac{0^2}{25} + \frac{3^2}{9} = 0 + 1 = 1$. ✓

Teacher Note

All four table points should be recognized as the axis intercepts of the ellipse — this bridges directly to the lesson's main idea. If students are confused by Q3, ask: "Did you recognize these points from Q2?" The verification step builds fluency with substituting parametric outputs into implicit equations, which is the core skill of EK 4.7.A.2.